Evaporation at a Liquid Surface Due to Jet Impingement

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Experiments were performed to determine mass transfer coefficients for evaporation from a water surface on which an air jet impinged. During the course of the experiments, parametric variations were made of the jet velocity and diameter, the separation distance between the jet origin and the water surface, the diameter of the water surface, and the degree of insulation of the water containment pan. It was found that for all of the investigated operating conditions, the dimensionless mass transfer coefficient varied with the 0.8 power of the jet Reynolds number. Furthermore, the transfer coefficient decreased linearly as the separation distance between the jet origin and the water surface increased, with the most significant decreases occurring at relatively small values of the surface-to-jet diameter ratio. At larger diameter ratios, the transfer coefficient was relatively insensitive to the separation. In general, the larger the diameter of the water surface, the lower the transfer coefficient. Comparisons with the literature showed that the dimensionless mass transfer coefficients for impingement on a liquid surface are lower than those for impingement on a solid surface.

Introduction

Jet impingement provides heat transfer coefficients that are among the highest available in any forced convection flow. This characteristic has motivated experimental study of a variety of jet/impingement-surface configurations, involving both heat transfer and mass transfer. The mass transfer work was limited to jet impingement on solid surfaces, and the resulting mass transfer coefficients were transformed to heat transfer coefficients by employing the well-established analogy between the two transport processes. A survey of the available heat and mass transfer literature on jet impingement has been provided by Martin [1].

The present work is concerned with evaporation mass transfer which occurs due to jet impingement on a liquid surface. The capability of the liquid surface to deform and to move in response to both the impinging jet and the ensuing wall jet gives rise to a pattern of fluid flow which differs from that existing adjacent to a solid surface. The response of the mass transfer coefficient to such an altered flow pattern will be investigated here by experiment.

The physical situation to be studied is the normal impingement of an air jet on a water surface. The jet is of circular cross section and impinges at the center of the water surface, which is also of circular planform. Measurement of the rate of evaporation and evaluation of the density of the water vapor at the liquid surface and in the jet from temperature and pressure measurements enabled the mass transfer coefficient and the Sherwood number to be determined.

The experiments encompassed systematic variations of four relevant physical parameters. These included the jet velocity, the diameters D_j and D_w of the jet and the water surface, and the separation distance S between the origin of the jet and the surface. In dimensionless terms, the parameters can be represented by the jet-origin Reynolds number Re and the dimension ratios D_w/D_j and S/D_j. The D_w/D_j ratio extended from 7.7 to 23.5 in six steps, with one of the D_w/D_j values being attained using two different combinations of D_w and D_j. This duplication was employed with a view to verifying the universality of D_w/D_j as a correlating parameter. The S/D_j ratio ranged from 5 to 15 in three steps. For each D_w/D_j, S/D_j combination, the jet Reynolds number covered the range from about 3800 to 15,000. The presentation of results will be structured so as to highlight the effect of each of the dimensionless parameters on the Sherwood number.

The densities of the water vapor at the liquid surface and in the jet were constant during a given data run but varied appreciably during the duration of the experimental work. Such variations were useful in that they provided the opportunity to demonstrate that the mass transfer coefficients were universal, i.e., independent of the densities of the water vapor. To vary the vapor density at the liquid surface, different degrees of insulation of the water containment pan were used. The response of the Sherwood number to the thus-achieved changes in vapor density will be highlighted. Density changes also occurred in response to climatic changes, i.e., changes in the relative humidity of the air.

Experimental Apparatus and Procedure

Apparatus. To achieve the well-defined thermal, vapor concentration, and hydrodynamic conditions necessary for the attainment of accurate and reproducible data, the experiments were performed in an environment chamber and in the suction mode. In this mode of operation, air from the temperature-controlled laboratory room was drawn into and through the chamber by a downstream-positioned blower, the discharge of which was vented outside the building. This arrangement avoided the uncontrolled compression-related preheating of the air and the hydrodynamic disturbances which would have occurred had the system been operated in the blowing mode, i.e., with an upstream-positioned blower.

The description of the experimental apparatus is facilitated by reference to Fig. 1. The environment chamber was a large, rectangular, metal-walled enclosure, completely airtight except for a vertically oriented jet delivery tube that pierced its upper wall and an air exit port that was centered in its lower wall. The chamber was 1.52 m in height, with a square horizontal cross section of dimensions 0.61×0.61 m. Only the upper portion of the chamber is depicted in Fig. 1.

The water to be evaporated was contained in a specially fabricated, straight-sided circular pan supported from below by a cylindrical column. Two such pans were employed during the course of the experiments, with respective water-surface diameters D_w of 16.34 and 22.30 cm (surface area ratio = 1.84) and a sidewall height of 2.54 cm. The pans were fabricated from 0.0635-cm-thick galvanized steel. Distilled water was used in all of the experiments.

The actual support of the pan was accomplished by three

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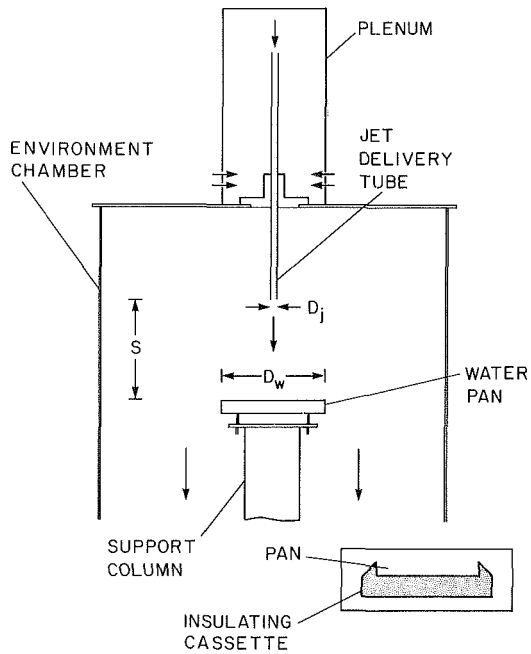

Fig. 1 Main features of the experimental apparatus

sharp-tipped Teflon screws threaded into a plexiglass flange at the top of the support column. Adjustment of the screws facilitated the leveling of the pan. The support column was a thin-walled 10.16-cm-dia pvc tube. Note that all the components of the support system were made of low-conductivity materials which, coupled with the point contact between the screw tips and the bottom of the pan, served to minimize extraneous heat conduction.

In a supplementary set of experiments, the water containment pan was enveloped at the sides and from below by an insulating cassette as pictured schematically in the inset at the lower right in Fig. 1. The presence of the insulation gave rise to lower temperatures of the evaporating water surface and to lower vapor pressures at the surface. The cassette was fabricated from closed-pore polystyrene insulation (Styrofoam) and covered with plasticized contact paper. Its outboard, upfacing surfaces were beveled downward at 45 deg in order to avoid altering the pattern of fluid flow adjacent to the water surface. Support of the cassette-enveloped pan was by the sharp-tipped Teflon screws as before.

The jet which impinged on the water surface originated at the exit cross section of a circular tube (i.e., the jet delivery tube). As seen in the figure, the delivery tube was supported and positioned by a sleevelike guide which rested in a lap in the upper wall of the environment chamber. The tube could be moved vertically within the guide, so that its exit could be positioned at any desired distance S from the water surface. At the lower end of the tube, the outer wall was beveled back from the exit cross section at 45 deg to minimize possible effects of the finite wall thickness.

Three jet delivery tubes were used throughout the course of the experiments, with respective internal diameters D_j equal to 0.950, 1.572, and 2.129 cm. A different custom-fitted guide was used with each tube. The lengths of the tubes were selected to ensure the attainment of hydrodynamically developed flow at the exit, and the respective length-diameter ratios of the tubes were 80, 60, and 50. The motivation for the attainment of developed flow is that such flows are characterized by well-defined and reproducible velocity distributions and turbulence levels.

The portion of the jet delivery tube which protruded above the upper wall of the environment chamber was enclosed in a cylindrical plenum chamber. The wall of the plenum was pierced near its lower end by an array of holes, and it was through these holes that air was drawn into the system. This plenum arrangement was adopted to ensure the accurate measurement of the initial state of the airflow by a psychrometric unit situated atop the upper wall of the environment chamber.

In the lower portion of the environment chamber (not shown in Fig. 1) was situated a balance for mass measurement and a platform. Immediately preceding and immediately following each data run, when measurements were made of the mass of the pan and its contents, the lower end of the support column rested on the weighing surface of the balance. During the data run proper and at all other times, the column rested on the platform. In the transfer of the support function from the balance to the platform, both the column and the balance remained stationary while the platform was slid under the flanged lower end of the column. The balance and the platform were positioned so as to avoid blocking the air exit port situated at the center of the lower wall of the environment chamber.

The environment chamber was equipped with two leakproof access windows. One of these was situated in the upper portion of the chamber and was used for filling and emptying the water pan and for positioning the pan and the jet delivery tube. The second window, situated in the lower part of the chamber, was used to facilitate the mass measurement operations and in transferring the support function from the platform to the balance and vice versa.

The upstream plenum, the jet delivery tube, and the environment chamber were part of an open-loop flow circuit. The air exiting the environment chamber was ducted through a flowmeter (a calibrated orifice plate) and a control valve to the blower, which was situated in a service corridor outside the laboratory.

Instrumentation. The humidity of the air entering the environment chamber was measured by a specially designed psychrometric unit. It consisted of a 76-cm length of 7.6-cm-dia plexiglass tube equipped at its downstream end with a fan which operated in the suction mode. The fan and the tube

Nomenclature

A = area of water surface $= \pi D_w^2/4$
D_j = diameter of jet at origin (i.d. of delivery tube)
D_w = diameter of water surface
$\mathfrak{D}$ = mass diffusion coefficient
h = enthalpy per unit mass
K = average mass transfer coefficient, equation (1)
ΔM = change of mass
$\dot{m}_j$ = rate of mass flow in jet delivery tube
p = pressure
Re = jet-origin Reynolds number $= 4\dot{m}_j/\mu\pi D_j$
S = separation distance between jet origin and water surface
Sc = Schmidt number
Sh = Sherwood number $= KD_w/\mathfrak{D}$
T_{db} = dry-bulb temperature
T_{wb} = wet-bulb temperature
T_{wtr} = water temperature
W = specific humidity
μ = viscosity
ν = kinematic viscosity
ρ = density
$\rho_{wv,j}$ = water vapor density in jet
$\rho_{wv,s}$ = water vapor density at liquid surface
τ = duration of data run

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cross-sectional area were chosen to yield an airflow velocity which provides near equality of the psychrometric wet-bulb temperature and the thermodynamic wet-bulb temperature. Wet- and dry-bulb thermometers, which could be read to 0.1°F or better, were inserted radially into the tube through cork apertures in the cylindrical wall. The thermometer bulbs were positioned at the tube centerline, with the dry-bulb thermometer upstream of the wet-bulb thermometer (the separation distance was 30 thermometer diameters). Radiation shielding was provided by aluminum foil which completely covered the outside wall of the plexiglass tube.

The air temperature in the environment chamber was sensed by a thermocouple situated at a distance of ½S above the water surface and displaced to the side of the jet by a distance of ¼D_w from the jet centerline. The thermocouple reading was virtually indistinguishable from that of the dry-bulb thermometer of the psychrometric unit.

Four thermocouples were used to measure the temperature of the evaporating water. One of these was situated at the centerline, and a second was positioned adjacent to the outer rim (about 0.3–0.4 cm from the rim). The other two thermocouples were installed so as to give a uniform radial spacing for the four thermocouples. Owing to the presence of surface motions which occurred for certain operating conditions, the thermocouple junctions were located about 0.6 cm below the nominal surface plane of the water. That placement ensured that the junctions were not exposed to the airflow. All thermocouples were made from precalibrated, Teflon-coated, 30-gage chromel and constantan wire.

Mass measurements were made with an Ohaus triple-beam balance having a capacity of 2610 g and which could be read to 0.05 g. The mass evaporated during a data run typically fell in the range of 20–50 g.

The pressure in the environment chamber was read relative to ambient by a Baratron solid-state, capacitance-type pressure meter having a resolution of 10^{-3} Torr. The same meter was used for the pressure measurements relevant to the orifice flowmeter. A barometer situated adjacent to the environment chamber provided the ambient pressure.

Procedure. It is well established that the temperature of an evaporating water surface is lower than that of the ambient air into which the evaporation takes place. Furthermore, if the water is initially at ambient temperature, a waiting period must be allowed for the water temperature to decrease to its equilibrium value for steady-state evaporation. In the present experiments, it was determined from preliminary data runs that equilibration periods of 1 and 1½ hours, respectively, for the smaller and larger pans, were more than sufficient to yield steady-state evaporation. Prior to the initiation of the equilibration period, the pan was overfilled to compensate for the expected loss of mass due to evaporation. During the period, the jet velocity and separation distance were those of the upcoming data run.

At the conclusion of the equilibration period, the pan remained slightly overfilled (an elevation of the surface of 0.05–0.1 cm at the center). This degree of overfilling ensured that the water level did not drop perceptibly below the upper edge of the pan sidewall during the ensuing data run. The termination of the equilibration period was immediately followed by the measurement of the mass of the pan and its contents, after which the data run proper was initiated.

During the run, the four thermocouples situated in the pan were read periodically – a total of twenty times. The wet- and dry-bulb thermometers, the environment chamber thermocouple, and the pressures were read four times at equal intervals. The duration of the run was chosen to yield evaporation-induced changes of mass in the 20 and 40 g ranges, respectively, for the smaller and larger pans. The mass was measured immediately after the termination of the run.

Data Reduction

The mass transfer coefficient K for the evaporation process was evaluated from the definition

$$K = (\Delta M/\tau A)/(\rho_{wv,s} - \rho_{wv,j}) \tag{1}$$

In this equation, ΔM is the measured change of mass which occurred during a time interval τ, while A is the area of the circular planform of the water surface (i.e., $A = \pi D_w^2/4$). The denominator is the difference between the densities of the water vapor at the water surface and in the jet, $\rho_{wv,s}$ and $\rho_{wv,j}$, respectively. Note that for the conditions of the experiments,

$$\rho_{wv,j} = \rho_{wv,\infty} \tag{2}$$

where $\rho_{wv,\infty}$ is the vapor density in the ambient within the environment chamber. From the standpoint of practice, $\rho_{wv,\infty}$ is the most convenient representation of $\rho_{wv,j}$, since it is well defined and can be readily determined.

For the evaluation of $\rho_{wv,s}$, it was assumed that phase equilibrium prevailed at the liquid–vapor interface (i.e., at the surface of the liquid), so that the vapor was saturated vapor corresponding to the water temperature T_{wtr}. The value of T_{wtr} was obtained by averaging the thermocouple readings both radially and temporally. Owing to the virtual uniformity of the radial temperature distribution, it was not necessary to weight the individual thermocouple readings. Then, in terms of standard notation (g = saturated vapor)

$$\rho_{wv,s} = \rho_g(T_{wtr}) \tag{3}$$

The evaluation of $\rho_{wv,j}$ begins with the determination of the specific humidity W of the air entering the upstream plenum from the laboratory. From any standard thermodynamics text,

$$W = [W_{wb} h_{fg}(T_{wb}) + c_{pa}(T_{wb} - T_{db})]/[h_g(T_{db}) - h_f(T_{wb})] \tag{4}$$

Here, h_f and h_g denote the enthalpies of saturated liquid and saturated vapor, and h_{fg} is the latent heat, each evaluated at the indicated temperature, where T_{db} and T_{wb} respectively denote the dry-bulb and wet-bulb temperatures measured by the psychrometric unit. In addition, c_{pa} is the specific heat of dry air (taken as 1.006 kJ/kg – °C) and W_{wb} is the specific humidity corresponding to the wet-bulb saturation condition, i.e.,

$$W_{wb} = 0.622 p_{\text{sat}}(T_{wb})/[p_{\text{bar}} - p_{\text{sat}}(T_{wb})] \tag{5}$$

in which p_{bar} and p_{sat} are, respectively, the barometric pressure in the laboratory and the saturation pressure of water vapor.

In the impinging jet, the specific humidity is equal to the value obtained from equation (4), but the pressure is no longer equal to p_{bar}. If Δp is the pressure drop between the laboratory and the environment chamber, the pressure in the jet can be expressed as $p_j = (p_{\text{bar}} - \Delta p)$. Then, in the jet,

$$W = 0.622 p_{wv,j}/(p_j - p_{wv,j}) \tag{6}$$

or

$$p_{wv,j} = W p_j/(0.622 + W) \tag{7}$$

from which the numerical value of $p_{wv,j}$ follows directly. Finally, from the definition of the relative humidity ϕ

$$\phi = p_{wv,j}/p_{\text{sat}}(T_j) = \rho_{wv,j}/\rho_g(T_j) \tag{8}$$

where the temperature T_j is the measured temperature in the environment chamber, which was virtually identical to the dry-bulb temperature at the psychrometric unit. Equation (8) can be solved for the numerical value of $\rho_{wv,j}$, which is then introduced into equation (1), thereby completing its evaluation.

All of the thermodynamic properties needed in the foregoing calculations were taken from the ASHRAE Handbook [2].

The mass transfer coefficients determined from equation (1) were recast in dimensionless form in terms of the Sherwood number

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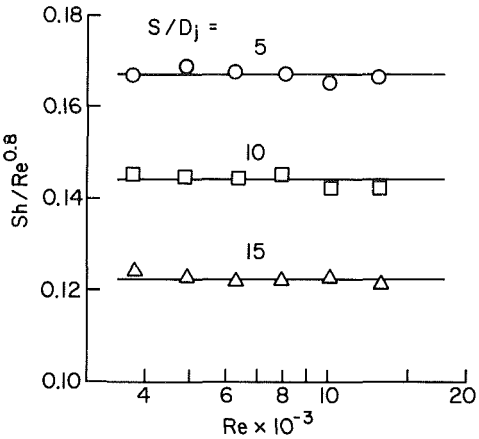

Fig. 2 Sherwood number results for $D_w/D_j = 7.68$

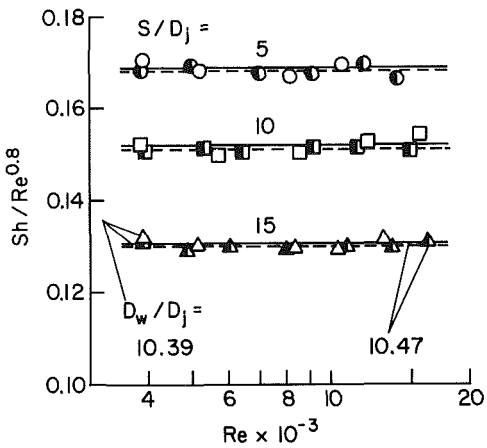

Fig. 3 Sherwood number results for $D_w/D_j = 10.4$

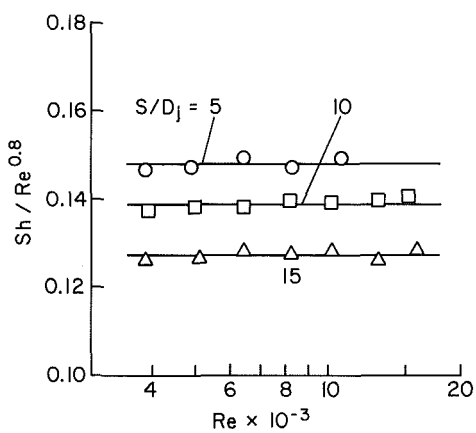

Fig. 4 Sherwood number results for $D_w/D_j = 14.18$

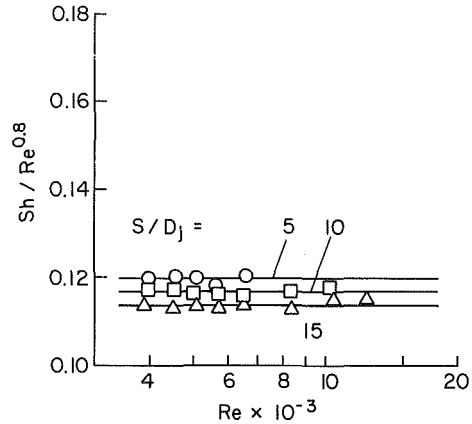

Fig. 5 Sherwood number results for $D_w/D_j = 17.20$

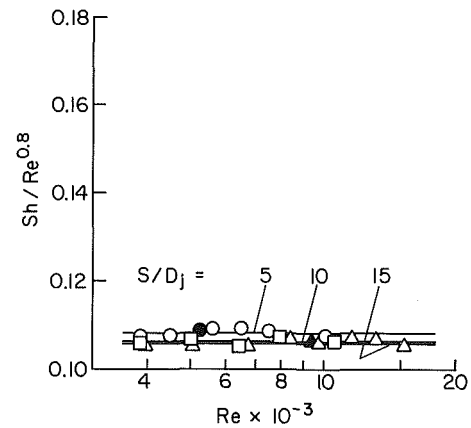

Fig. 6 Sherwood number results for $D_w/D_j = 23.47$

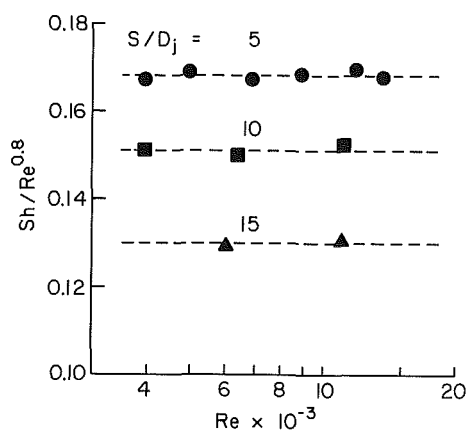

Fig. 7 Sherwood number results for the insulated water containment pan, $D_w/D_j = 10.47$

$$\mathrm{Sh} = KD_w/\mathfrak{D} \tag{9}$$

By introducing the Schmidt number $\mathrm{Sc} = \nu/\mathfrak{D}$, the mass diffusion coefficient $\mathfrak{D}$ can be eliminated from equation (9), so that

$$\mathrm{Sh} = (KD_w/\nu)\mathrm{Sc} \tag{10}$$

For the air/water vapor system, $\mathrm{Sc} = 0.6$ (Table 10.1 of [3] and Table 7 of [4]).

The kinematic viscosity ν $(=\mu/\rho)$ appearing in equation (10) was evaluated at jet conditions. The viscosity μ of mixtures of air and water vapor is not known to high precision, but for the jet temperatures and vapor concentrations of the present experiments, it did not differ by more than 0.2 percent from that for dry air [4]. In view of this, dry-air viscosities were used. The density of the air/water vapor mixture comprising the jet is given by

$$\rho_j = \rho_{wv,j} + \rho_{a,j} \tag{11}$$

where $\rho_{wv,j}$ is known from equation (8) and $\rho_{a,j}$ is determined by using the perfect gas law with T_j and $p_{a,j}$ $(=p_j - p_{wv,j})$ as inputs.

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The Sherwood number results were parameterized by the Reynolds number Re at the jet origin (i.e., at the exit of the jet delivery tube)

$$\mathrm{Re} = 4\dot{m}_j/\mu\pi D_j \tag{12}$$

in which $\dot{m}_j$ is the rate of mass flow through the delivery tube. The value of $\dot{m}_j$ was obtained from the measured flow rate at the orifice meter by subtracting the rate of evaporation at the water surface.

Results and Discussion

An analysis of the data indicated that for a given geometric configuration characterized by fixed values of D_w/D_j and S/D_j, the Sherwood number varied with the 0.8 power of the Reynolds number. This suggests a generalized presentation of results in terms of $\mathrm{Sh}/\mathrm{Re}^{0.8}$ rather than a presentation of Sh itself.

The $\mathrm{Sh}/\mathrm{Re}^{0.8}$ data for all of the investigated operating conditions are plotted in Figs. 2–7. Each figure pertains to a specific value of the diameter ratio D_w/D_j of the water surface and the jet, starting with $D_w/D_j = 7.68$ in Fig. 2 and increasing from figure to figure to $D_w/D_j = 23.47$ in Fig. 6. Among the six values of D_w/D_j that were employed to parameterize the experiments, two were nearly identical, namely, 10.39 and 10.47. The results for both these cases are presented in Fig. 3 to facilitate their direct comparison. In addition, results from supplementary experiments in which the water containment pan was insulated are presented in Fig. 7.

In each of Figs. 2–7, the $\mathrm{Sh}/\mathrm{Re}^{0.8}$ ratio is plotted as a function of the Reynolds number Re for three parametric values of the dimensionless separation distance S/D_j between the jet origin and the water surface, namely, $S/D_j = 5$, 10, and 15. The horizontal lines that pass through the data for each S/D_j are least-squares fits. To facilitate figure-to-figure comparisons, the ordinate scales for all of the figures were made identical.

An overall inspection of Figs. 2–7 indicates that the deviations of the data from the $\mathrm{Sh} \sim \mathrm{Re}^{0.8}$ correlation do not exceed about two percent, and there is no evidence of systematic deviations with either D_w/D_j or S/D_j. The 0.8 power strongly suggests that the airflow passing over the water surface is turbulent, and the early onset of turbulence is believed due to the presence of roughnesslike disturbances and waves on the surface. This matter will be elaborated later when comparisons are made with the literature on jet impingement on solid surfaces.

In general, for fixed values of the water surface diameter and the jet diameter (i.e., fixed D_w/D_j), the mass transfer coefficient decreases as the separation between the jet origin and the water surface increases. From a comparison of the successive figures, it is seen that the sensitivity of the transfer coefficient to the separation distance is markedly affected by the D_w/D_j ratio. Over all, there is a trend for the coefficient to become less and less dependent on the separation distance as D_w/D_j increases. This is especially strongly manifested for the three largest investigated D_w/D_j (14.18, 17.20, and 23.47). In particular, at $D_w/D_j = 23.47$, the effect of S/D_j is only 2–3 percent throughout its entire range. This insensitivity is physically plausible since the details of the impingement which are affected by the separation distance (i.e., impingement velocity and size of impingement zone) are relatively unimportant when the impingement zone is very much smaller than the overall area of the surface.

Further comparisons of the successive figures reveal an initial increase of $\mathrm{Sh}/\mathrm{Re}^{0.8}$ with D_w/D_j (compare Figs. 2 and 3), followed by a marked decrease at still larger values of D_w/D_j. In rationalizing these results, it is convenient to think in terms of a fixed jet diameter and, thereby, to associate increases in D_w/D_j with increases in the diameter of the water surface. Thus, the wall jet which is spawned by the impinging jet traverses a longer radial distance along the water surface as D_w/D_j increases, with related increases in the thicknesses of the velocity and vapor-concentration boundary layers. These thicker boundary layers give rise to lower mass transfer coefficients for larger D_w/D_j. The Sherwood number, however, contains the product of the mass transfer coefficient and the diameter of the water surface, which vary oppositely with D_w/D_j. At smaller D_w/D_j, the two opposing factors are competitive, leading to the initial increase of the Sherwood number, while at larger D_w/D_j, the decrease of the mass transfer coefficient dominates.

Attention will now be turned to Fig. 3 and to the comparison of results for two cases having virtually identical values of the D_w/D_j ratio but for which the individual values of D_w and of D_j differed. These cases were investigated with a view to exploring the universality of D_w/D_j as a correlation parameter. The data corresponding to $D_w/D_j = 16.34/1.572 = 10.39$ are represented by the half-black symbols and the dashed least-squares lines, while those for $D_w/D_j = 22.30/2.129 = 10.47$ are represented by the open symbols and the solid least-squares lines.

As seen in the figure, the data for the two cases are nearly coincident. In fact, the maximum deviation of the corresponding least-squares correlating lines is 0.6 percent. This excellent agreement provides strong testimony in support of the universality of D_w/D_j as a correlation parameter.

As noted earlier, auxiliary experiments were undertaken to demonstrate that the Sherwood number results are independent of the densities of the water vapor. To supplement the naturally occurring changes in vapor density which accompanied climatic changes (i.e., changes in the relative humidity of the ambient air), more controlled changes of the vapor density at the liquid surface were achieved by the use of the insulating cassette described earlier. The presence of the insulation brought about a decrease in the water temperature which, in turn, decreased the vapor pressure and vapor density at the surface. For instance, for one set of operating conditions, the values of $(T_{wtr} - T_{wb})/(T_{db} - T_{wb})$ were 0.321 and 0.243, respectively, without and with the insulation. The corresponding values for another case were 0.208 and 0.134.

The Sherwood number data obtained with the insulating cassette in place are represented by the black data symbols in Fig. 7 and by the pair of black data points in Fig. 6. In Fig. 7, the dashed lines are the least-squares lines corresponding to the uninsulated containment pan, i.e., the same dashed lines that are plotted in Fig. 3. Both Figs. 7 and 6 display excellent agreement between the Sherwood numbers for the uninsulated and insulated pans. Since the two sets of results correspond to different vapor densities at the liquid surface, it may be concluded that the Sherwood numbers are universal with respect to this parameter. Furthermore, from the fact that the Sherwood numbers were also independent of the vapor densities in the jet, it follows that they are universal with respect to both the vapor densities at the surface and in the jet.

Consideration will now be given to quantifying the variations of $\mathrm{Sh}/\mathrm{Re}^{0.8}$ with D_w/D_j and S/D_j. These variations have been discussed qualitatively in connection with Figs. 2–6, but in those figures both D_w/D_j and S/D_j served as discrete parameters rather than as continuous independent variables, so that their quantitative influences were not readily discernible.

In Fig. 8, $\mathrm{Sh}/\mathrm{Re}^{0.8}$ is plotted as a function of D_w/D_j for parametric values of S/D_j. The data points shown in the figure correspond to the least-squares horizontal lines in Figs. 2–6 and are, therefore, representative of the entire investigated Reynolds number range. Smooth curves have been passed through the data for continuity.

The figure shows that the overall variation of the Sherwood number with D_w/D_j at a fixed Reynolds number is substantial

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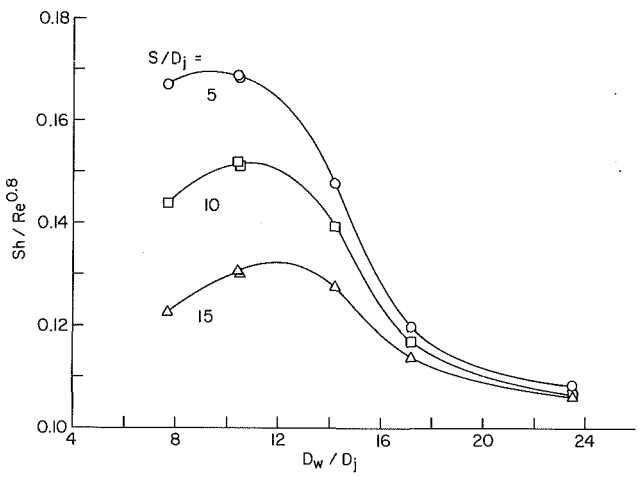

Fig. 8 Variation of the Sherwood number with D_w/D_j

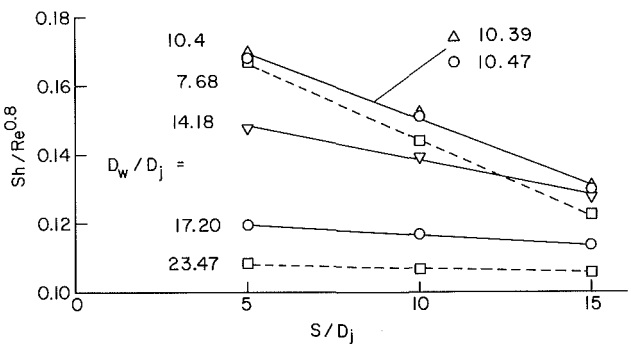

Fig. 9 Variation of the Sherwood number with S/D_j

when the separation distance between the jet origin and the water surface is relatively small. For example, for $S/D_j = 5$, the minimum and maximum values of $\mathrm{Sh}/\mathrm{Re}^{0.8}$ are 1.083 and 1.695. As the separation distance increases, there is a marked decrease in the variation of $\mathrm{Sh}/\mathrm{Re}^{0.8}$ with D_w/D_j, so that for $S/D_j = 15$, $\mathrm{Sh}/\mathrm{Re}^{0.8}$ is confined to the range between 1.058 and 1.323. Notwithstanding the just-documented fact that the overall variation of Sh with D_w/D_j is quite sensitive to S/D_j, the sensitivity dies away at larger D_w/D_j, where all the results tend to merge.

All of the curves in Fig. 8 are characterized by a common shape, which includes an initial rise, the attainment of a maximum, and a sharp dropoff which becomes increasingly more gradual at larger D_w/D_j. As noted earlier, the maximum in the Sh versus D_w/D_j distribution should not be regarded as indicating a maximum value of the mass transfer coefficient. Rather, the maximum in Sh is due to the conflict between K and D_w. Indeed, it is readily shown that the mass transfer coefficient decreases monotonically with D_w/D_j over the investigated range.

Figure 9 presents the variation of $\mathrm{Sh}/\mathrm{Re}^{0.8}$ with S/D_j, with D_w/D_j as the curve parameter. Again, the data points appearing in the figure correspond to the least-squares horizontal lines in Figs. 2–6. As expected, the mass transfer coefficient drops off as the separation distance increases.

The figure shows that the variation of $\mathrm{Sh}/\mathrm{Re}^{0.8}$ with S/D_j at a fixed value of D_w/D_j is well represented by a straight line. The slope of the straight line is markedly affected by the D_w/D_j ratio. At small values of D_w/D_j, the transfer coefficient is quite sensitive to the separation distance, while at large D_w/D_j, it is virtually independent of the separation distance.

With regard to comparisons with the literature, it was already noted in the Introduction that prior jet-related mass transfer experiments appear to be limited to impingement on solid surfaces. In the absence of prior liquid-surface impingement results, the present Sherwood numbers will be compared with those for impingement on a solid surface.

An empirical correlation of Sherwood number data for jet impingement on solid surfaces was presented in [1]. That correlation, when rephrased in terms of the present notation and evaluated for $\mathrm{Sc} = 0.6$, is

$$\mathrm{Sh} = 1.614F(\mathrm{Re})[(D_w/D_j) - 2.2]/[(D_w/D_j) + 0.2((S/D_j) - 6)] \tag{13}$$

where $D_w/D_j > 5$. For the Reynolds number range between 2000 and 30,000, which includes the range investigated here,

$$F(\mathrm{Re}) = 1.36\mathrm{Re}^{0.574} \tag{14}$$

At higher Reynolds numbers, the exponent increases, attaining a value of 0.775 for Re between 120,000 and 400,000.

The $\mathrm{Re}^{0.574}$ and $\mathrm{Re}^{0.8}$ dependences of the Sherwood number, respectively, for solid-surface impingement and liquid-surface impingement, provide clear evidence of the major differences in the transfer processes which prevail in the two situations. The solid-surface case is characterized by laminar flow over a smooth, flat surface. On the other hand, the liquid-surface case corresponds to turbulent flow over a compliant surface which deforms in response to the airflow and whose deformation, in turn, affects the airflow pattern. Careful visual observations made during the course of the experiments revealed a depression of the water surface in the impingement zone, with the size and depth of the depression depending on the Reynolds number and on the separation distance. In the region away from the impingement zone, surface waves were in evidence. From the standpoint of the airflow, these waves acted like major roughness elements, in all likelihood causing flow separation. At Reynolds numbers higher than those for which data were collected, the relatively deep depression caused by the impinging jet tended to be unstable, and the surface waves were large and irregular.

It is believed that both the surface depression in the impingement zone and the waves situated in the region away from the impingement zone contribute to diminished rates of mass transfer. This conjecture is substantiated by comparisons between the present data and equations (13)–(14). Such comparisons show that the Sherwood numbers for impingement on a liquid surface are consistently lower than those for impingement on a solid surface.

Concluding Remarks

The experimental work reported here is, seemingly, the first study of evaporation from a liquid surface due to jet impingement. With water as the evaporating liquid and ambient air as the impinging jet, parametric variations were made of the jet velocity and diameter, the separation distance between the jet origin and the water surface, and the diameter of the water surface. In dimensionless terms, these parameters can be represented by S/D_j, D_w/D_j, and Re, where S is the separation distance, D_w and D_j are the water surface and jet diameters, and Re is the jet-origin Reynolds number. The investigated ranges of these groups extended from 5 to 15 for S/D_j, from 7.68 to 23.47 for D_w/D_j, and from 3800 to 15,000 for Re. The experiments yielded mass transfer coefficients, which were presented in terms of the Sherwood number Sh based on the water surface diameter.

It was found that for all of the investigated operating conditions, the Sherwood number varied with the 0.8 power of the Reynolds number. Furthermore, at a fixed Reynolds number and D_w/D_j ratio, the Sherwood number decreased linearly as the separation distance between the jet origin and the water

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surface increased. The decrease was significant at small values of D_w/D_j, but, at large values of D_w/D_j, the Sherwood number was virtually independent of the separation distance.

The mass transfer coefficient decreased monotonically with increases in the diameter of the water surface (although the Sherwood number displayed a local maximum). The overall change of the coefficient in response to changes in the water surface diameter was greatest at small separation distances.

Supplementary experiments involving different jet diameters and different water surface diameters, but with a common value of D_w/D_j, affirmed the universality of the D_w/D_j ratio as a correlation parameter. Other experiments demonstrated that the Sherwood number results were independent of the densities of the water vapor at the liquid surface and in the impinging jet.

Comparisons with the literature showed that the Sherwood numbers for jet impingement on the liquid surface are lower than those for jet impingement on a solid surface.

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